DESCRIPTION OF COURSEWORK

Course Code	EEE312
Course Name	Programming Practices
Lecturer	Hamid Tahaei
Academic Session	2022/09
Assessment Title	Assignment 2

A. Introduction/ Situation/ Background Information

This assignment applies the programming techniques learned on solving linear equation.

B. Course Learning Outcomes (CLO) covered

At the end of this assessment, students are able to:

CLO 1 Design program using advanced algorithms and techniques

CLO 2 Use modern programming techniques to solve scientific problems

C. University Policy on Academic Misconduct

1. Academic misconduct is a serious offense in Xiamen University Malaysia. It can be defined as any of the following:
   i. **Plagiarism** is submitting or presenting someone else's work, words, ideas, data or information as your own intentionally or unintentionally. This includes incorporating published and unpublished material, whether in manuscript, printed or electronic form into your work without acknowledging the source (the person and the work).
   ii. **Collusion** is two or more people collaborating on a piece of work (in part or whole) which is intended to be wholly individual and passed it off as own individual work.
   iii. **Cheating** is an act of dishonesty or fraud in order to gain an unfair advantage in an assessment. This includes using or attempting to use, or assisting another to use materials

that are prohibited or inappropriate, commissioning work from a third party, falsifying data, or breaching any examination rules.

2. All the assessment submitted must be the outcome of the student. Any form of academic misconduct is a serious offense which will be penalised by being given a zero mark for the entire assessment in question or part of the assessment in question. If there is more than one guilty party as in the case of collusion, both you and your collusion partner(s) will be subjected to the same penalty.

D. Instruction to Students

1. This is an individual assignment.

2. Answer the questions in **Section F Tasks**.

3. Submission deadline is 13th December 2022.

4. Submit your C codes for the two questions to MOODLE. Name your C source files as "<your_student_ID>_Assignment2.c". Also, Submit the answer cover page paper with your signature and screen shot of your code for each question and the output. Rename the assignment cover page to "<your_student_ID>.pdf". A Written report is not required. However, you are required to provide explanation for each block/ line of code using the comment operator such as // in the C file.

E. Evaluation Breakdown

No.	Component Title	Percentage (%)
1.	Report	8
2.	C Code Question 1	7
	TOTAL	**15**

F. Task(s)

1. Write a C program to solve the following linear equations using the **Gauss-Seidel** method:

$$\begin{bmatrix} 2+3i & -i & 0 \\ -i & 3-i & i \\ 0 & i & 2-2i \end{bmatrix} \begin{bmatrix} x_0 \\ x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 1-2i \\ -i \\ 0 \end{bmatrix}$$

Your answers for $x_0, x_1 and\ x_2$ are complex quantities. Stop the iteration when the quantity $max = \left(\left|x_i^{(k)} - x_i^{(k+1)}\right|\right) < 0.000001$.

[3 marks]

2. Write a C program to find the differences of the roots found by Bisection method and the Fixed-point iteration method for the following function:

$$f(x) = 3x^3 - 6x + 1 \ for \ 0 < x < 1$$

Start your iteration with the proper initial value x0 to converge the solution to the root. Stop the iteration when the quantity $|x_i - x_{i-1}| < 0.000001$.

Print out the value xi and the error at every *i-th* iteration.

[4 marks]

APPENDIX 1

MARKING RUBRICS

Component Title	Assignment 1				Percentage (%)	15	
Criteria	Score and Descriptors					Weight (%)	Marks
	Excellent (41-50)	Good (31-40)	Average (21-30)	Need Improvement (11-20)	Poor (0-10)		
Short Report	Very detailed description of program flow, algorithm and computational aspects of the problem. Excellent explanation of numerical methods applied to solve the engineering problem.	Good description of program flow and algorithm used. Able to explain numerical methods to solve the engineering problem.	Sufficient information on program flow but lacking details description. Able to explain numerical methods used to solve the problem but not in details.	Missing important information on program flow and algorithm. Descriptions contain minor errors. Explanation of numerical solutions is not sufficient.	Not sufficient information on the program flow and algorithm used. Serious technical error in the description of the program. Unable to explain numerical methods applied to the problem.	**8**	
C Code	Program works According to Given requirements of The questions. Able to apply Numerical methods learned to solve the problem. Creative use of algorithm and/or advanced programming techniques to improve computational efficiency. Use good programming practices and Code optimization for writing easy to understanding code and efficiency.	Program works according to the given the requirements of the question. Able to apply numerical methods learned to solve the problem. Organization of the code can be improved to simplify understanding and improve execution efficiency.	Program works but only fulfilling partial requirements of the questions. Able to apply numerical methods learned to solve the problem. Code is difficult to understand, not optimized for efficiency.	Program works but results are not correct. Only part of the requirements of the question are fulfilled. Unable to apply numerical methods learned to solve the problem.	Program does not work. No results are produced. Unable to apply numerical methods learned to solve the problem.	**7**	
					TOTAL	**15**	

Note to students: Please print out and attach this appendix together with the submission of coursework